# Premankur Banerjee

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My research lies at the confluence of Haptic perception and multimodal feedback, Extended Reality (VR/AR/MR), and embodied human–AI interaction. I design, develop and evaluate end-to-end interactive systems that integrate real-time sensing, feedback, and new forms of physical interaction (direct and AI-mediated), to study how people perceive and engage with these systems. My work focuses on transforming prototypes into validated, scalable systems, turning bold ideas into innovative, tangible technologies and *impossible* interactions part of everyday life.

## EDUCATION

**University of Southern California** Aug 2023 – Present
Ph.D. in Computer Science; GPA: 3.95/4.00
Expected graduation: **Summer 2027**.

**Technical University of Munich** Oct 2019 – Dec 2021
M.Sc. in Electrical and Computer Engineering, *with distinction*; GPA: 1.4/1.0

**University of Calcutta** Aug 2015 – May 2019
B.Tech. in Electronics and Communications Engineering; GPA: 9.16/10.00

## APPOINTMENTS

**DropIn – *Stealth Startup*** Oct 2026 – Present
*Chief Technology Officer (CTO)*
Los Angeles, CA

**University of Southern California – HaRVI Lab** Aug 2023 – Present
*Graduate Research Assistant, Haptics Robotics and Virtual Interactions Lab*
Thomas Lord Department of Computer Science, Los Angeles, CA

**Sorbonne Université – ISIR** Jun 2022 – May 2023
*Research Engineer, Human–Computer Interface Group*
Institut des Systèmes Intelligents et de Robotique, Paris, France

**German Aerospace Center (DLR)** May 2021 – Nov 2021
*Research Intern, Thesis*
Munich, Germany

**Technical University of Munich – LMT** Nov 2019 – Apr 2021
*Research Intern, Chair of Media Technology*
Munich, Germany

**Indian Space Research Organization – RRSC, ISRO** Apr 2018 – Aug 2018
*Research Intern, Remote Sensing Group*
Kolkata, India

## FELLOWSHIPS, RESEARCH SUPPORT, AND HONORS

### Fellowships and Scholarships

**Link Foundation Fellowship** in Modeling, Simulation, and Training ($35,000) 2026–2027

**USC Viterbi School of Engineering Fellowship** ($42,000) 2025–2026

**Deutschlandstipendium** (German National Scholarship), TU Munich (€7,200) 2020–2021

### Research Support

Southern California Clinical and Translational Science Institute (NIH/NCATS) Grant ($100,000) 2023–2024

### Awards

**Undergraduate Mentor Award**, Thomas Lord Department of Computer Science, USC Viterbi School of Engineering 2025

## PUBLICATIONS

### Peer-Reviewed Journal Articles

[J1] **P. Banerjee**, V. Chen, J. Lee, and H. Culbertson. Virtual Bodies, Real Touch: Haptic Social Communication Across VR, MR, and Cross-Reality. **Frontiers in Virtual Reality**, vol. 7, 2026.

[J2] **P. Banerjee***, N. Li*, J. Wang, and H. Culbertson. *Shaping Redirected Walking with Ankle-Based Vibrotactile Feedback*. **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, presented at IEEE ISMAR 2026.

[J3] **P. Banerjee***, S. Sweidan*, C. Yunis, V. H. Cruz, and H. Culbertson. *Crip Haptic Jam*: Vibrotactile Composition for Disability-Centered Dance. **IEEE Transactions on Haptics** (2026), special issue on Haptics for Arts.

[J4] **P. Banerjee**, M. Montiel, L. Tomita, O. Means, J. Kutch, and H. Culbertson. The Impact of Airflow and Multisensory Feedback on Immersion and Cybersickness in a VR Surfing Simulation. **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, presented at IEEE VR 2025.

### Journal Manuscripts Under Review

[R1] **P. Banerjee** and H. Culbertson. Human-in-the-Loop Calibration of Propulsion-Driven Locomotion in Multisensory Virtual Reality. Submitted to **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, 2026.

### Peer-Reviewed Conference Papers

[C1] **P. Banerjee**, J. Wang, O. Means, and H. Culbertson. SWIMVR: Simulating Water Interaction Using Multimodal Vibro-Thermal Rendering. **IEEE Haptics Symposium**, 2026.

[C2] **P. Banerjee**, J. Wang, L. Tomita, M. Montiel, and H. Culbertson. Virtual Encounters of the Haptic Kind: Towards a Multi-User VR System for Real-Time Social Touch. **IEEE World Haptics Conference (WHC)**, 2025.

[C3] **P. Banerjee**, J. Cherin, J. Upadhyay, J. Finley, J. Kutch, and H. Culbertson. Perception and Control of Surfing in Virtual Reality Using a 6-DoF Motion Platform. **EuroHaptics**, 2024.

[C4] H. Singh, L. P. Marcilla, **P. Banerjee**, M. Rothammer, and T. Hulin. FerroVibe: Towards a Modular Tactile Device. **IEEE Conference on Telepresence**, 2024.

[C5] **P. Banerjee**, E. Muschter, H. Singh, B. Weber, and T. Hulin. Towards a VR Evaluation Suite for Tactile Displays in Telerobotic Space Missions. **IEEE Aerospace Conference**, 2023.

[C6] G. Bailly, C. Alberge, S. Haliyo, P. Gauthier, **P. Banerjee**, A. Baudry, and S. Hadjes. Le Raffinement d'un Système Haptique pour la Réalité Virtuelle. In Proceedings of the **36th Conference on l'Interaction Humain-Machine (IHM)**, 2025.

### Conference Manuscripts Under Review

[R2] **P. Banerjee**, X. Yu, K. Lee, M. Mozeh, and H. Culbertson. *Shifting from Human Intent to AI Authorship across Three Loci of Agency in Mediated Social Touch.* Submitted to **ACM CHI 2027**.

### Theses

[T1] Premankur Banerjee. ViESTac: A Multimodal VR Evaluation Suite for Novel Tactile Devices. Master's thesis, Technical University of Munich; DLR-IB-RM-OP-2021-214, 104 pages, 2021.

### Demonstrations

[D1] H. Singh, T. Hulin, and **P. Banerjee**. Rendering and Displaying Tactile Properties of Virtual Objects via ViESTac and FerroVibe. **IEEE World Haptics Conference**, 2023.

## PATENT APPLICATIONS

[P1] Heather Marie Culbertson, Premankur Banerjee, Jason James Kutch, Evelyne Falann Bernardette Morisseau. **A Virtual Reality Motion Platform**. PCT application **PCT/US2026/041074**; USC Ref. 2025-099-02; MCC Ref. 11760-012WO1. Filed August 13, 2026.

## RESEARCH EXPERIENCE AND SELECTED PROJECTS

**University of Southern California – HaRVI Lab** **2023–Present**

**Surfing in Virtual Reality** *Project Lead*

Developed a VR surfing simulator integrating a 6-DoF motion platform, center-of-pressure-based navigation, and multisensory feedback. Conducted user studies investigating motion perception, control, and user experience.

**Multi-User Virtual System for Bidirectional, Mediated, Real-Time Social Touch** *Project Lead*

Developed a multi-user VR/MR social-touch system using 52-actuator wearables and low-latency wireless communication, facilitating bidirectional physical interaction and studying presence, embodiment, and affective experience.

**AI-Mediated Affective Haptic Interactions** *Project Lead*

Developed an AI-mediated social-touch framework combining LLM-based haptic generation, multimodal affect sensing, and real-time feedback across human-controlled, human-prompted, and autonomous interaction.

**Redirected Walking in VR Using Ankle-Based Haptics** *Project Co-Lead*

Developed a gait-synchronized ankle-worn haptic system with wireless vibrotactile actuation and IMU sensing.

**Disability-Centered Vibrotactile System** *Project Co-Lead*

Collaborated with USC School of Cinematic Arts, the Leonardo Crip Tech Incubator, and ASU to co-design modular wearable haptics and a Unity-based touch composition interface with Disabled artists, evaluated through user studies and live performances.

**Sorbonne Université – ISIR (HCI Group)** **2022–2023**

Developed real-time force-feedback control for a room-scale, encounter-type haptic device delivering over 100 N of kinesthetic feedback. Investigated full-body VR interactions, redirected walking, and visuo-haptic illusions.

**German Aerospace Center (DLR)** **2021**

Developed a device-agnostic multimodal VR evaluation framework for comparing tactile displays across friction, temperature, texture, and shape modalities, validated through user studies with two novel tactile devices.

**Technical University of Munich – LMT** **2020–2021**

Developed autoencoder-based vibrotactile signal compression for networked haptic communication. Implemented ROS–Unity robotic simulations like autonomous grasping and liquid-pouring tasks for Kinova MOVO in Gazebo.

## TEACHING EXPERIENCE

**CSCI-526: Advanced Mobile Devices and Game Consoles** Fall 2026
*Teaching Assistant*
University of Southern California.

**DSCI-517: Research Methods and Analysis for User Studies** Fall 2024
*Teaching Assistant*
University of Southern California.

**EI72071: Computational Haptics Laboratory** Summer 2021
*Lab Lecturer and Teaching Assistant*
Technical University of Munich.

**EI5052: Time-Varying Systems and Computation** Winter 2021
*Teaching Assistant*
Technical University of Munich.

**EI7341: Image and Video Compression** Spring 2021
*Teaching Assistant*
Technical University of Munich.

**EI70220: Digital Signal Processing** Summer 2020
*Teaching Assistant*
Technical University of Munich.

## RESEARCH MENTORING

**Jiaxuan Wang** (USC CS master student) | now: **RayNeo Innovation Technology, Shenzhen**,

**Evelyne Morisseau** (UC Berkeley MechE undergrad, SURE program) | now: **PhD student at Cornell**,

**Vivien Chen** (USC CS undergrad) | now: **SWE Intern, Meta**,

**Kyuhong Lee** (USC CS undergrad) | now: **SWE Intern, Google**,

**Mia Montiel** (USC BME undergrad) | now: **Medical student at MCW**,

**Olivia Means** (USC BME undergrad, AME master student) | now: **HRL Laboratories**,

**Lauren Tomita** (USC BME undergrad) | now: **Medtronic**,

**Arike Cooker** (USC BME undergrad) | now: **White Memorial Medical Center**,

**Xander Neary** (UCLA CS undergrad, SURE program), **Brandon Xu** (USC CS master student),

**Joanne Lee** (USC CS undergrad), **Joyce Ng** (USC CS undergrad), **Magen Mozeh** (USC CS PDP),

**Arushi Mundergi** (USC CS undergrad), **Chelsea Zhang** (USC CS undergrad), **Olivia Xue** (USC CS PDP),

**Pablo Murillo** (USC CS VSI undergrad), **Deborah Folorunsho** (USC CS VSI undergrad),

**Mayte Morales Ramirez** (USC CS VSI undergrad), **Chitkala Dharmavaran** (USC CS master student)

Read some of the student testimonials here.

## INVITED RESEARCH TALKS

**Indian Institute of Technology (IIT) Bombay**, hosted by Dr. Suhas Kakade, India — Dec 2025

**USC Computer Science Department Advisory Board**, hosted by Prof. Nenad Medvidović, USA — Nov 2025

**University of Calcutta**, Institute of Radio Physics and Electronics (IRPE), India, hosted by Prof. Sumitra Mukhopadhyay — Jul 2024

**TU Dresden**, Centre for Tactile Internet with Human-in-the-Loop (CeTI) Convention, Germany — Feb 2022

## ACADEMIC AND PROFESSIONAL SERVICE

### Peer Reviewing – Journals

International Journal of Human–Computer Interaction; IEEE Transactions on Haptics (ToH); IEEE Transactions on Visualization and Computer Graphics (TVCG) IEEE Robotics and Automation Letters (RA-L); Frontiers in Virtual Reality; Springer Nature – *Virtual Reality*; Perspectives of Communication & Media (PCM); Wisdom Academic Press.

### Peer Reviewing – Conferences

ACM Conference on Human Factors in Computing Systems (CHI) 2027; IEEE World Haptics Conference (WHC) 2023, 2025; IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) 2025; IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2026; ACM Symposium on User Interface Software and Technology (UIST) 2025; IEEE Haptics Symposium 2024, 2026; EuroHaptics 2022, 2024;

## MEDIA COVERAGE AND SCIENTIFIC OUTREACH

### Research and Press Coverage

- Multi-user mediated social-touch work featured by USC Today[1], Interesting Engineering (TikTok), YouTube, and the Arduino blog.
- VR surfing research featured by KTLA, USC Today[2], USC Today[3], USC Viterbi Magazine, and USC Health Sciences Campus News.
- *Crip Haptic Jam* featured by the Leonardo Crip Tech Incubator and Ability Central.

### Scientific Outreach

- Volunteered to teach Robotics to children at Peace4Kids in Los Angeles.
- Demonstrated CoVR at Fête de la Science 2022, contributing to public engagement in science and technology.

## TECHNICAL EXPERTISE

- **Programming and Software:** C++, C, C#, Python, MATLAB, Unity 3D, Unreal Engine, CHAI3D, OpenHaptics, Arduino and microcontroller programming, Blender, FreeCAD, ROS.
- **Haptics, XR, and Sensing:** Actuators, force sensing/feedback, tactile sensing, wearables, filtering, temporal smoothing, multiplayer VR/AR, pressure sensing, affective haptics, gesture input, motion/body tracking (OptiTrack, Intel RealSense, MediaPipe, BlazePose), 6-DoF motion platforms, telepresence, generative haptics, mediated social touch.
- **Simulation and Robotics:** MuJoCo, Gazebo, physics modeling, OpenGL, room-scale VR, autonomous robotics, inverse kinematics, IMUs, sensor fusion, signal processing, real-time networking (TCP/UDP), wireless protocols, teleoperation.
- **Machine Learning:** CNNs, autoencoders, interactive machine learning, large language models, prompt engineering, local LLM inference, multimodal affect recognition, affective computing, PyTorch, TensorFlow, Transformers.
- **Human-Subjects Research:** Experimental design, user-centered design, protocols, data processing, and statistical analysis (ANOVA, nonparametric tests, mixed models); studies involving more than 200 participants in total.

## ADDITIONAL INFORMATION

**Languages:** English, Bengali, Hindi (native); German (professional); French and Spanish (elementary).

**Hobbies:** Acting, filmmaking, VFX design, reading, photography, singing, and film scoring.